Reading group notes: linear attention

Reading group of the Ashbury machine learning lab

1 Background

A practical system has to balance the cost of the key dimension against the quality of the outer-product write. This work studies linear attention, a method that rethinks the decay gate. We argue that the causal mask deserves as much attention as the read-out projection. Despite steady progress, the interaction between the associative recall and the recurrent state is still poorly understood.

Our analysis covers models up to 1.3B parameters, and larger models may behave differently. Combining linear attention with methods that target the outer-product write is a promising direction. We expect the state size to interact with the feature map at larger scales, which we leave to future work. The read-out projection also makes linear attention easier to inspect, since its behaviour can be read off per token.

Recent work has made the recurrent state a central concern, because it decides how far linear attention models can scale. Despite steady progress, the interaction between the outer-product write and the chunk-wise kernel is still poorly understood. Most prior methods treat the state size as fixed and tune the chunk-wise kernel around it. A practical system has to balance the cost of the causal mask against the quality of the read-out projection.

2 Methods we read

Gradients through the outer-product write are clipped at a norm of 5.0. During inference linear attention caches the associative recall, so each new token costs a constant amount of work. The causal mask in linear attention is parameterised by a small linear layer with 256 hidden units. Unlike softmax attention, linear attention updates the state size only when the decay gate changes.

Linear attention computes the recurrent state once per layer and reuses it for every head. A coefficient controls how strongly the associative recall depends on the recurrent state; we set it to 0.5 throughout. Unlike softmax attention, linear attention updates the associative recall only when the read-out projection changes. The decay gate is initialised so that linear attention starts training close to gated linear attention.

During inference linear attention caches the feature map, so each new token costs a constant amount of work. The recurrent state is initialised so that linear attention starts training close to a Transformer++ baseline. The outer-product write in linear attention is parameterised by a small linear layer with 32 hidden units. To keep training stable, linear attention normalises the associative recall before it is combined with the causal mask.

3 Results we discussed

Variance across seeds stays below 0.4 points for every configuration we report. Ablating the feature map lowers the WikiText score by 6.7 points, the largest drop among all components. Linear attention scores 81.4% on LAMBADA, compared with 50.6% for gated linear attention. The gains of linear attention grow with the input length, as the key dimension becomes the bottleneck.

We train every model for 80B tokens with the same data order and report the mean of three seeds. On LAMBADA, linear attention improves over softmax attention by 3.7 points. The gains of linear attention grow with the input length, as the key dimension becomes the bottleneck. Linear attention scores 76.7% on WikiText, compared with 64.6% for a Transformer++ baseline.

Variance across seeds stays below 0.2 points for every configuration we report. Linear attention scores 44.8% on MQAR, compared with 43.3% for gated linear attention. We train every model for 40B tokens with the same data order and report the mean of three seeds. With the associative recall frozen, linear attention still recovers most of its improvement on MQAR.

4 Open questions

The read-out projection also makes linear attention easier to inspect, since its behaviour can be read off per token. Our analysis covers models up to 1.3B parameters, and larger models may behave differently. The results suggest that the causal mask is the main lever for quality at a fixed budget. We expect the causal mask to interact with the decay gate at larger scales, which we leave to future work.

Our analysis covers models up to 1.3B parameters, and larger models may behave differently. We expect the decay gate to interact with the recurrent state at larger scales, which we leave to future work. The state size also makes linear attention easier to inspect, since its behaviour can be read off per token. A limitation of linear attention is that the key dimension must be tuned separately for each model size.